A MINI PROJECT REPORT

ON

SMART ACCIDENT DETECTION AND EMERGENCY RESPONSE SYSTEM

BACHELOR OF TECHNOLOGY

Computer Science & Engineering (AIML)

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CERTIFICATE

This is to certify that the mini project report entitled “Smart Accident Detection and Emergency Response System” has been prepared by Ambuj Kumar (2501921530056), Aditi Srivastava (2505111530007), Dhriti Singh (2505111530077), and Akansha Chaurasia (2501921530037), Department of Computer Science and Engineering (AIML), G.L. Bajaj Institute of Technology & Management, Greater Noida, under the supervision of Dr. Santosh Kumar Srivastava, Associate Professor.

The report presents an academic prototype and design study. Any statement of implementation or performance should be read in the context of the evidence and limitations described in the report. This work has not been submitted earlier for the award of another degree or diploma, to the best of the authors’ knowledge.

Date	Supervisor Signature	Dean / Head of Department
_____	_____	_____

ACKNOWLEDGEMENT

The completion of this mini project reflects guidance, discussion, experimentation, and repeated refinement. The team expresses sincere gratitude to Dr. Santosh Kumar Srivastava, Associate Professor, Department of CSE (AIML), for supervision and academic direction. The team also thanks the department and institute for providing an environment in which a practical safety-related problem could be explored through engineering methods.

The project benefited from conversations about sensor behavior, application workflow, data handling, interface clarity, and the importance of documenting limitations. Appreciation is extended to classmates and peers who offered feedback during planning and review. Any remaining errors or shortcomings are the responsibility of the project team.

“A project becomes stronger when its authors remain willing to question their first design and learn from what the evidence reveals.”

Ambuj Kumar • Aditi Srivastava • Dhriti Singh • Akansha Chaurasia

ABSTRACT

Road accidents create an urgent need for dependable information flow between a suspected incident and those who may provide assistance. This project proposes a Smart Accident Detection and Emergency Response System that connects motion sensing, candidate-event analysis, confirmation, location capture, structured incident creation, and a responder dashboard. The objective is to reduce avoidable information gaps while making the status of an alert visible and auditable.

The proposed architecture combines an ESP32 and MPU6050 hardware exploration with an application stack specified as Android/Kotlin, Spring Boot REST APIs, MySQL, and a React/HTML/CSS responder dashboard. The design distinguishes sensor input from event classification and distinguishes server acceptance from responder acknowledgment. It also considers duplicate alerts, missing or stale location, network failures, security, privacy, and human factors.

The report expands the project into requirements, architecture, implementation planning, test strategy, results interpretation, limitations, and future work. It emphasizes that sensor spikes alone cannot establish that a collision occurred. Any operational deployment would require empirical calibration, representative datasets, security testing, controlled field evaluation, and authorized integration with emergency services. The report therefore presents a grounded academic prototype rather than claiming certified emergency readiness or proven reductions in response time or fatalities.

Keywords Accident detection, ESP32, MPU6050, GPS, IoT, emergency notification, incident management, Spring Boot, MySQL, responder dashboard, safety engineering.

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LITERATURE SURVEY AND DESIGN CONTEXT

Prior work has explored the use of smartphone sensors and contextual information to identify possible road accidents and notify responders. The supplied project material references Thompson et al. (2010), White et al. (2011) on WreckWatch, and Fernandes et al. (2015). These studies motivate combining sensing, location, and notification, while also highlighting the need to manage false positives and contextual uncertainty. Full bibliographic details should be verified against the original publications before final institutional submission.

The research gap adopted for this project is integration across the complete workflow: detection, verification, precise and fresh location, structured alert, duplicate handling, and responder coordination. This is a project-level synthesis, not a claim that no prior system has addressed any individual component.

Study / direction	Relevance	Design lesson
Smartphone sensing	Uses motion/context signals to identify candidate events.	Sensors require contextual interpretation and false-alarm controls.
WreckWatch-style workflow	Combines detection, GPS, and notification.	A useful alert needs location and supporting context.
Multimodal notification	Combines vehicle or mobile data with communication.	End-to-end delivery and acknowledgment need separate states.

> *“The gap is not simply detecting an event; it is carrying trustworthy information from uncertainty to accountable action.”*

REFERENCES AND EVIDENCE POLICY

The literature survey and motivation draw on the references named in the supplied mini-project presentation. The following entries preserve the source-level details available in that material; exact volume, issue, page, DOI, and publisher metadata should be checked against the original publications before formal submission.

[1] C. Thompson et al. (2010). “Using Smartphones to Detect Car Accidents and Provide Situational Awareness to Emergency Responders.” MOBILWARE.

[2] J. White et al. (2011). “WreckWatch: Automatic Traffic Accident Detection and Notification with Smartphones.” Mobile Networks and Applications.

[3] B. Fernandes et al. (2015). “Mobile Application for Automatic Accident Detection and Multimodal Alert.” IEEE VTC Spring.

[4] Ministry of Road Transport and Highways, Government of India. *Road Accidents in India 2023.* Source named in the supplied presentation; consult the original report for official tables and definitions.

[5] World Health Organization. *Global Status Report on Road Safety 2023.* Source named in the supplied presentation; consult the original publication for global estimates and methodology.

Methodology and evidence policy. The report uses a design-oriented approach: define stakeholder needs, convert them into requirements, propose a layered architecture, and organize validation into component, integration, usability, reliability, and security tests. Evidence should be labelled as proposed, simulated, bench-tested, integrated-tested, or not tested. A diagram or mock screen is not proof of a live emergency-service integration.

> *“A responsible report makes the boundary between what was designed, what was observed, and what remains unknown easy to see.”*

CHAPTER 1

1.1 Introduction

Road safety is not only a transport problem; it is a time-critical coordination problem. When a collision occurs, useful assistance depends on recognizing a possible event, establishing whether it is genuine, communicating where it happened, and ensuring that the information reaches someone able to act. The proposed Smart Accident Detection and Emergency Response System treats these actions as one connected workflow rather than separate tasks. The report describes an academic prototype and its intended design; it does not claim a certified or production-ready emergency service.

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CHAPTER 1

1.2 Context and Need

A person involved in a serious crash may be disoriented, injured, or unable to make a call. Nearby witnesses may not know the correct location or what details responders need. An automated system can reduce some of this friction by preparing a structured incident record. Automation should support human judgment, not replace dispatchers, clinicians, police, or established emergency channels.

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CHAPTER 1

1.3 Problem Definition

The core problem is the delay and information loss that can occur between a suspected road incident and organized assistance. A useful solution must balance sensitivity with false-alarm control, location accuracy with privacy, and fast notification with verification. A single sensor spike is not enough to conclude that a collision has occurred. The system therefore frames detection as a candidate-event decision followed by confirmation and controlled notification.

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CHAPTER 1

1.4 Project Overview

The project combines a sensing concept, a decision workflow, GPS coordinates, a structured alert, and a responder-facing dashboard. The project material specifies Android/Kotlin for a mobile workflow, Spring Boot REST APIs for service communication, MySQL for persistent records, and a React/HTML/CSS dashboard. The hardware exploration includes an ESP32 board, MPU6050 motion sensor, and buzzer. These elements represent the planned prototype architecture and should be validated individually.

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CHAPTER 1

1.5 Existing System and Gaps

Manual emergency calls rely on a person being able to communicate and on a caller knowing the location. Smartphone-based detection can add accelerometer, gyroscope, GPS, and context, but sensors can confuse potholes, drops, sudden braking, and collisions. Research projects demonstrate parts of the pipeline; integration remains difficult when alert quality, location freshness, responder acknowledgment, and duplicate suppression are treated separately.

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CHAPTER 1

1.6 Proposed System

The proposed workflow receives sensor or test data, calculates an event score, requests confirmation when feasible, captures location and timestamp, creates a structured incident, and displays it to authorized responders. A real emergency message should be sent only through approved and tested channels. The dashboard is intended to help a responder review location, time, severity indicators, and status without claiming to diagnose injury or determine the final medical response.

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Conceptual workflow; live emergency integration requires validation and authorization.

Figure 1.6 Proposed System — conceptual design view.

CHAPTER 1

1.7 Objectives

The objectives are to define a modular accident-management workflow; describe functional and non-functional requirements; examine sensor and software integration; propose a transparent decision process; design an incident data model; and evaluate the prototype through controlled test cases. Secondary objectives include reducing avoidable information gaps, preserving a clear audit trail, and documenting limitations honestly.

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CHAPTER 1

1.8 Scope

The academic scope covers requirements, architecture, data design, interface concepts, event logic, and test planning. It can demonstrate simulated incidents and bench-level sensor behavior. It does not establish that the system has been validated on public roads, connected to official dispatch, or proven to reduce injury or mortality. Those outcomes require permissions, field trials, ethics review where applicable, and collaboration with relevant authorities.

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CHAPTER 1

1.9 Stakeholders

Primary stakeholders include vehicle occupants, bystanders, emergency responders, hospitals, system administrators, and academic evaluators. Each stakeholder has different needs: the occupant needs a clear confirmation experience; the responder needs accurate and concise incident data; the administrator needs security and auditability; and the evaluator needs reproducible tests and transparent limitations. Design choices should make these trade-offs visible.

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CHAPTER 1

1.10 Unique Features

The distinctive design emphasis is the end-to-end workflow: candidate detection, confirmation, location capture, structured notification, responder status, and incident logging. A dashboard can make the state of an alert visible, while duplicate detection can reduce repeated notifications. These are design intentions to be tested rather than claims of demonstrated field performance.

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CHAPTER 1

1.11 Impact Rationale

The value of an accident-response system is not measured by the number of sensors or lines of code. Its value lies in whether it improves the quality and timeliness of decisions under stressful conditions. A well-designed alert may help a responder orient quickly; a poor alert may distract, mislead, or waste scarce attention. The system must therefore be judged by both helpful actions and preventable harms.

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> *“A useful alert is not merely fast; it is accurate enough to help, clear enough to act on, and restrained enough not to create avoidable harm.”*

CHAPTER 1

1.12 Chapter Summary

This chapter establishes the problem, project scope, objectives, stakeholders, and design philosophy. The central principle is that automated detection should initiate a disciplined process, not issue an unquestioned verdict. Subsequent chapters translate this principle into requirements, architecture, implementation planning, evaluation, and future work.

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CHAPTER 2

2.1 Requirement Analysis Overview

Requirements convert a broad social need into observable system behavior. For this project, each requirement should be traceable to a stakeholder need and to a test case. Requirements are written in terms of what the system must do, while design decisions explain how it may do it. Keeping these separate reduces the risk of treating a preferred technology as if it were the actual objective.

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CHAPTER 2

2.2 Functional Requirements

The system should acquire sensor or test input, estimate whether an event merits review, allow confirmation or cancellation where appropriate, capture location and timestamp, create an incident record, issue an authorized notification, show incident status, and maintain an audit trail. The precise priority and severity categories must be defined with domain experts before any real-world deployment.

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ID	Requirement	Acceptance evidence
FR1	Acquire sensor or test input	Timestamped valid sample or explicit error
FR2	Evaluate candidate event	Score or rule output recorded
FR3	Support confirmation	Confirm/cancel/timeout state visible
FR4	Capture location	Coordinates, timestamp and quality state
FR5	Create incident	Unique ID and persistent record
FR6	Notify workflow	Delivery state explicit
FR7	Update dashboard	Authorized incident view refreshes

CHAPTER 2

2.3 Data Requirements

A minimum incident record may include a unique identifier, event time, location coordinates, source of the event, confidence or rule score, confirmation status, alert status, and timestamps for status changes. Personal identifiers should be minimized. The data model should distinguish raw sensor readings from derived scores and should preserve whether a field was measured, inferred, or manually entered.

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Data group	Example fields	Control
Incident	ID, created time, status	Unique ID, allowed state changes
Location	Latitude, longitude, accuracy	Validate range and freshness
Sensor summary	Peak magnitude, sample window	Record source and units
Confirmation	User choice, timeout	Audit state and timestamp
Delivery	Queued, accepted, acknowledged	Do not collapse states
Audit	Actor, action, time	Role control and retention

CHAPTER 2

2.4 Performance Requirements

Performance requirements should be measurable and tested on target devices and networks. Relevant dimensions include event-to-screen latency, API response time, location freshness, successful record creation, dashboard update delay, and resource consumption. Values should be set after prototype measurement rather than invented in advance. An emergency system must report uncertainty and degraded operation rather than silently appearing healthy.

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Measure	How to measure	Reporting rule
Detection latency	Input timestamp to candidate score	Report median and high percentile
API latency	Request to response	Repeat under stated network conditions
Dashboard delay	Record commit to visible row	Separate server and client delays
Location freshness	Event time minus location time	Show missing/stale status
False-alarm burden	False alerts per operating hour	Use labelled test set
Availability	Successful checks over test period	Define observation window

CHAPTER 2

2.5 Reliability and Availability

A reliable design considers what happens when GPS is unavailable, mobile data drops, a sensor disconnects, or a server returns an error. Local buffering may preserve an event for later transmission, but delayed alerts must be labelled as delayed. Recovery behavior should be tested explicitly. The system must not suggest that an emergency message was delivered when only a local record was created.

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CHAPTER 2

2.6 Security and Privacy

Incident location is sensitive. Access should be restricted by role, transport should be encrypted in a deployed system, and logs should avoid unnecessary personal data. Retention rules must define how long records remain available and who may export them. A security design should also consider spoofed coordinates, repeated submissions, stolen devices, and unauthorized dashboard access.

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CHAPTER 2

2.7 Usability Requirements

The interface should work under stress, with concise language, clear status labels, large touch targets, and a confirmation window that does not obstruct urgent action. Error messages should say what happened and what the user can do next. Colour must not be the only way to communicate severity; labels and icons should reinforce the same meaning.

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CHAPTER 2

2.8 Data Quality

Location quality depends on signal availability, accuracy, and age. Sensor quality depends on device mounting, calibration, sampling rate, and noise. Each incident should therefore retain data-quality indicators. A location point with poor accuracy should not be presented as exact, and a missing value should not be replaced by a fabricated default.

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CHAPTER 2

2.9 Use-Case Analysis

The occupant or sensing device initiates a candidate event; the system evaluates it; the user may confirm or cancel; the incident service records the event; and an authorized responder reviews and updates its status. An administrator manages accounts and system configuration. Emergency services are external actors and must not be represented as integrated unless a real, authorized interface has been established.

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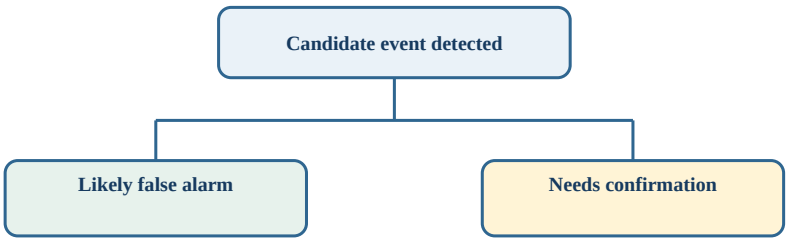


Figure 2.9 Use-Case Analysis — conceptual design view.

CHAPTER 2

2.10 SDLC Model

An iterative model is appropriate because sensor thresholds, interface wording, and alert workflow require repeated testing. Each cycle should define a small objective, implement it, test it, document results, and review risks. Early cycles can use simulated input; hardware integration and any field evaluation should come only after bench tests and safety review.

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CHAPTER 2

2.11 Acceptance Criteria

Acceptance criteria should state observable outcomes. For example, a valid test event creates one unique record; a duplicate within the configured window is linked or suppressed; missing GPS is represented explicitly; an API failure is visible; and a responder status change is recorded with time and user identity. These criteria are more meaningful than a vague statement that the system “works.”

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> *“In a safety-related system, an untested requirement is an intention—not evidence.”*

CHAPTER 2

2.12 Requirement Traceability

A traceability matrix links requirement IDs to design modules, implementation tasks, and tests. It allows a reviewer to see whether a safety-related requirement has been tested and whether an implementation change creates a new risk. Traceability is particularly important where the prototype may later be extended toward real emergency communication.

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CHAPTER 3

3.1 System Design Introduction

System design separates the sensing layer, decision logic, application service, persistent storage, and responder interface. The separation makes the prototype easier to test and allows a sensor or dashboard to be replaced without rewriting every other component. Clear boundaries also expose where trust is required: sensor data may be noisy, network requests may fail, and dashboard users need authorization.

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CHAPTER 3

3.2 Design Approach

A modular, service-oriented design is proposed. The mobile client collects or receives inputs and communicates with a backend through documented API endpoints. The backend validates requests, assigns incident identifiers, persists records, and exposes authorized views. The dashboard consumes these views and sends status updates. The modules should exchange explicit data structures rather than rely on hidden shared state.

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CHAPTER 3

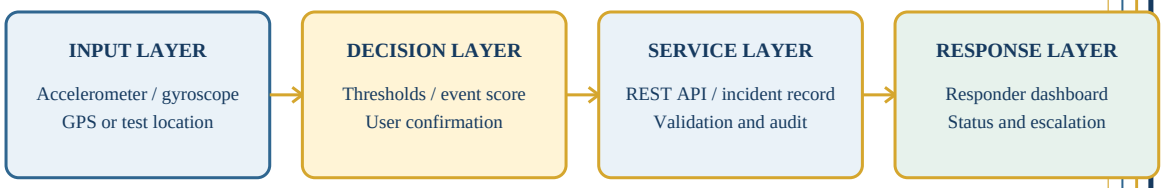
3.3 System Architecture

The logical architecture has four layers: input, decision, service, and response. Inputs include accelerometer and gyroscope readings, GPS, or controlled test events. Decision logic calculates a candidate score and confirmation requirement. The service layer validates and stores incident data. The response layer presents an incident to an authorized user and records acknowledgment or status changes.

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Logical architecture proposed in the project; deployment components must be tested independently.

Figure 3.3 System Architecture — conceptual design view.

CHAPTER 3

3.4 Sensor Subsystem

The MPU6050 combines a three-axis accelerometer and gyroscope. Acceleration can help identify abrupt movement; angular velocity can describe rotation. Neither directly proves a collision. Orientation, mounting position, vibration, and vehicle dynamics affect the signal. Calibration and repeated tests are required before choosing thresholds, and the system should use time windows rather than a single isolated sample.

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Signal	Possible value	Caution
Acceleration	Abrupt movement magnitude	Potholes and drops may look similar
Angular rate	Rotation and orientation change	Mounting affects axes
GPS	Approximate incident position	Can be stale or unavailable

CHAPTER 3

3.5 ESP32 Integration

The ESP32 can act as a microcontroller for bench-level acquisition and alert indication. The project material identifies ESP32 boards and an MPU6050 sensor, with a buzzer as a local output. Hardware validation should begin with power and ground checks, then I2C detection, then raw readings, and finally controlled motion tests. A successful sensor scan confirms bus communication, not correct accident classification.

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CHAPTER 3

3.6 Event-Decision Logic

A transparent baseline can combine acceleration magnitude, angular-rate magnitude, persistence over a short window, and contextual checks. The resulting score should be interpreted as a candidate-event indicator. A confirmation prompt, cooldown period, and duplicate-event logic can reduce unnecessary alerts. Thresholds shown in an academic diagram are illustrative until calibrated on representative data.

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CHAPTER 3

3.7 Location Subsystem

GPS coordinates should be paired with a timestamp and, where available, an accuracy estimate. The system should detect stale or missing coordinates and clearly show the age of the last valid location. A map marker can aid orientation, but responders need a textual address or landmark when available and should not assume that a coordinate always corresponds to a safe access point.

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CHAPTER 3

3.8 Backend API Design

The backend should expose narrow endpoints for creating an incident, retrieving authorized incident details, updating status, and checking service health. Each request needs validation, authentication, and useful error responses. The service should distinguish “accepted by server” from “delivered to responder” and “acknowledged by responder.” These are different states and should not be collapsed into one success message.

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CHAPTER 3

3.9 Database Design

A relational model can store incident, sensor summary, location, responder assignment, status history, and audit events. Foreign keys maintain relationships; indexes support queries by time, status, and incident identifier. Sensitive fields should be minimized and retention rules applied. A status-history table is preferable to overwriting every prior state because it preserves the sequence of response actions.

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Entity	Purpose	Key relation
Incident	Current incident summary	One-to-many status history
SensorSummary	Derived input summary	Belongs to incident
StatusHistory	Chronology of changes	Incident + actor
Responder	Authorized user record	Assignment relation

CHAPTER 3

3.10 Dashboard Design

The responder dashboard should prioritize active incidents, location, time, event source, confirmation status, and current response state. Secondary details can appear on selection. Filters should be predictable and accessible. Colour should be restrained: navy for structure, pale blue for neutral information, amber for pending attention, and red only for a defined high-priority state. Colour meanings require labels and should not imply clinical certainty.

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CHAPTER 3

3.11 Failure Handling

Failures should be explicit and recoverable. If the sensor disconnects, the acquisition state becomes unavailable. If GPS is missing, the alert indicates location unavailable or stale. If the API is unreachable, the client can show pending transmission without claiming delivery. If duplicate reports arrive, the service can flag possible duplicates while preserving an audit record.

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CHAPTER 3

3.12 Architecture Review

The design is modular and testable, but several dependencies remain open: real GPS source, device-to-server connectivity, identity management, deployment environment, notification provider, and authorized emergency-service integration. These should be recorded as assumptions and resolved before any field deployment. The architecture diagram describes intended boundaries, not proof that every integration has been completed.

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> *“The architecture is only as trustworthy as the weakest hand-off between detection, location, communication and acknowledgment.”*

CHAPTER 4

4.1 Implementation Overview

Implementation should follow the modular architecture and be recorded in small, reproducible increments. The supplied project stack specifies Android/Kotlin for the mobile workflow, Spring Boot for REST services, MySQL for persistent storage, and React with HTML/CSS for the responder dashboard. The ESP32 and MPU6050 belong to the hardware exploration. This chapter explains the implementation plan and validation approach without overstating unverified integration.

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Conceptual workflow; live emergency integration requires validation and authorization.

Figure 4.1 Implementation Overview — conceptual design view.

CHAPTER 4

4.2 Development Environment

A reproducible environment records operating system, compiler or runtime versions, library versions, board model, wiring map, and configuration values. The report should include these details when they are confirmed. Secrets, production credentials, and private keys must never be embedded in the report or source repository. Configuration should be separated from application logic.

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CHAPTER 4

4.3 Android/Kotlin Client

The mobile client can manage permission requests, sensor or test input, event confirmation, location acquisition, and alert status. Permission denial should be handled gracefully. The interface must not imply continuous tracking if tracking is not active, and should make it clear when a location is unavailable. User cancellation must have a defined meaning and must not erase audit data silently.

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CHAPTER 4

4.4 Spring Boot Service

The service layer validates incoming payloads, applies authorization, assigns unique IDs, persists incidents, and returns explicit response states. Request validation should reject malformed coordinates and impossible timestamps. API contracts should specify required fields, optional fields, error codes, and idempotency behavior. A repeated request should not accidentally create multiple emergency incidents.

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CHAPTER 4

4.5 MySQL Persistence

Database implementation should define table schemas, primary keys, foreign keys, indexes, and migration scripts. A sample schema can be tested with synthetic data. The database should not be treated as a substitute for backup, access control, or retention policy. Logs should support audit without duplicating sensitive information unnecessarily.

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CHAPTER 4

4.6 Responder Dashboard

The dashboard should show a compact queue of incidents and a detail view for a selected event. Status changes require authenticated roles and should be recorded with timestamps. A stale incident must be visibly different from a newly received one. Empty states, loading states, errors, and permission failures deserve as much design attention as the ideal success screen.

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CHAPTER 4

4.7 Hardware Wiring and Safety

For bench testing, the ESP32, MPU6050, power regulator, and buzzer should be wired according to board-specific pin documentation. A common ground is required for signal reference; I2C uses SDA and SCL pins supported by the chosen board. Power rails must be checked with a meter before connecting the sensor. USB power, external supply, and regulator output must not be combined without understanding the board circuitry.

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CHAPTER 4

4.8 Sensor Acquisition Code

The acquisition routine should initialize I2C, verify the sensor address, check device identity where supported, and report clear errors. It should collect timestamped readings and handle missing or invalid samples. Serial output is useful during bench tests but should not become the only monitoring mechanism. Sensor initialization success does not prove that the values are calibrated or that the detection model is valid.

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CHAPTER 4

4.9 API and Data Validation

Validation should occur at the client for user experience and again at the server for security and consistency. Coordinates must be in valid ranges; incident timestamps should be parseable; status transitions should follow an allowed sequence; and event identifiers should be unique. Server-side validation remains authoritative because client input can be altered or replayed.

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CHAPTER 4

4.10 Error Logging and Diagnostics

Logs should record diagnostic context such as module, event ID, error category, and timestamp. They should avoid passwords, access tokens, and unnecessary personal details. A useful diagnostic message identifies the failed stage and next check. For example, “sensor not detected on I2C bus” is more actionable than “system error.”

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CHAPTER 4

4.11 Coding Standards

Use meaningful names, small functions, consistent formatting, and comments that explain why a safety-related decision exists. Constants such as thresholds and cooldown periods should be configurable and versioned. Avoid embedding unverified assumptions in code. Unit tests should cover boundary values, missing data, duplicate submissions, and invalid state transitions.

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CHAPTER 4

4.12 Implementation Status and Evidence

The report must distinguish design artifacts, simulated demonstrations, bench-level hardware checks, and integrated end-to-end tests. Evidence can include dated screenshots, serial logs, test records, and API responses. A diagram or mock interface is not evidence of a live integration. The final submission should replace planned-work statements with verified results only where test evidence exists.

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> *“A prototype becomes credible not when it looks complete, but when its boundaries, failures and evidence are made visible.”*

CHAPTER 5

5.1 Evaluation Approach

Evaluation should begin with component-level checks and progress toward integrated tests. A useful sequence is sensor detection, stable readings, event scoring, confirmation handling, location capture, API persistence, dashboard display, and status update. Each test needs a precondition, input, expected outcome, observed outcome, and evidence reference. The report should not substitute expected behavior for an observed result.

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Stage	Pass condition
Component	Input and output observed reliably
Integration	Data travels across defined boundaries
Failure handling	Degraded state is explicit
Usability	Representative task completed without critical confusion

CHAPTER 5

5.2 Test Environment

The test environment should identify the board, sensor, wiring, power source, mobile device or simulator, backend version, database version, and network conditions. Synthetic events are appropriate for early tests because they are repeatable and safe. Any physical-motion tests should be controlled, low-risk, and conducted without endangering people or property.

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CHAPTER 5

5.3 Sensor Connectivity Tests

A sensor connectivity test verifies that the selected I2C address is visible and that the expected registers or library initialization succeed. It should test repeated power cycles and cable movement. A single successful scan is a useful diagnostic observation, but it does not establish stable readings or accurate event detection.

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CHAPTER 5

5.4 Event Classification Tests

Classification tests should include benign abrupt movements, ordinary travel vibration, stationary handling, simulated impact patterns, and missing samples. The test set should record true positives, false positives, true negatives, and false negatives where a trusted label exists. Threshold tuning on the same small dataset used for evaluation can overstate performance; a separate validation set is preferable.

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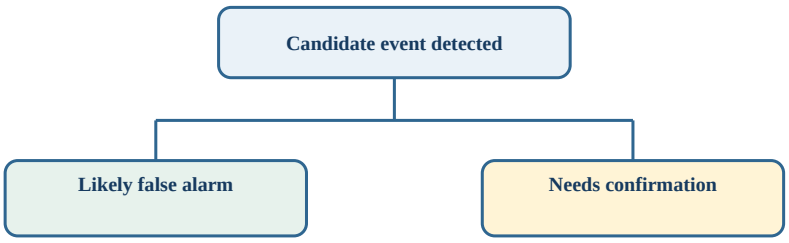


Figure 5.4 Event Classification Tests — conceptual design view.

CHAPTER 5

5.5 Confirmation Workflow Tests

Test whether the confirmation prompt appears when expected, whether cancellation is recorded, and whether a timeout produces the defined fallback. The fallback must be carefully designed: a timeout is not proof of an accident, but it may justify a different level of review if the system has been approved for that use. Timing values should be selected through usability and safety evaluation.

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CHAPTER 5

5.6 GPS and Timestamp Tests

Test valid coordinates, unavailable GPS, poor accuracy, stale location, and clock inconsistencies. The interface should display uncertainty and age. If the event is created without location, it should remain clearly marked rather than receiving a fabricated coordinate. A test should confirm that server and client timestamps are interpreted consistently.

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CHAPTER 5

5.7 Duplicate Alert Tests

Generate repeated reports from the same device and reports with close timestamps and nearby coordinates. Duplicate logic should consider a configurable time window and distance threshold while avoiding the suppression of genuinely separate incidents. Suspected duplicates should be linked or flagged rather than irreversibly discarded.

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CHAPTER 5

5.8 API and Database Tests

Test successful creation, malformed requests, unauthorized requests, repeated idempotency keys, database unavailability, and status transitions. Confirm that records persist after service restart and that error responses do not reveal sensitive internals. A successful HTTP response should be interpreted according to the API contract, not automatically as responder acknowledgment.

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CHAPTER 5

5.9 Dashboard Usability Tests

Ask representative users to find the newest incident, identify its location, distinguish pending from acknowledged status, and update a status. Observe errors and completion time. The evaluation should include accessibility considerations such as keyboard navigation, contrast, labels, and screen-reader support where applicable.

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CHAPTER 5

5.10 Performance Measurement

Measure latency at defined points: sensor event to candidate score, candidate score to incident creation, API request to database commit, and record creation to dashboard visibility. Report median and high-percentile values over repeated runs, together with network and device conditions. Do not publish a single best-case time as a general performance guarantee.

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CHAPTER 5

5.11 Results and Interpretation

Results should be presented with clear evidence labels: observed, simulated, planned, or not tested. If a test was not conducted, say so. If the sample size is small, explain that the result is exploratory. A meaningful discussion explains what the evidence supports, what it does not support, and what should be tested next.

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> *“Evidence is the difference between a promising story and a dependable system.”*

CHAPTER 5

5.12 Discussion of Impact

Potential impact includes earlier recognition of an incident, more structured location information, fewer duplicated notifications, and improved responder visibility. These are plausible benefits, not demonstrated public-health outcomes. Demonstrating impact requires comparison with a baseline, representative users, field-relevant conditions, and a careful analysis of false alarms and missed events.

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CHAPTER 6

6.1 Conclusion

The project defines a connected accident-management workflow that links sensing, candidate-event analysis, confirmation, location capture, structured incident creation, and responder-facing status. Its central contribution is the integration concept and the disciplined separation of stages. The report does not claim that the prototype is a certified emergency product or that it has proven reductions in response time or fatalities.

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CHAPTER 6

6.2 Key Contributions

The project organizes the problem into modular layers, identifies functional and non-functional requirements, proposes a data model, outlines a responder dashboard, and establishes a test strategy. It also emphasizes duplicate handling, location quality, auditability, and explicit failure states—issues that can be overlooked when a project focuses only on detecting a sudden movement.

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CHAPTER 6

6.3 Current Limitations

Sensor signals can be ambiguous; GPS may be unavailable or imprecise; network access can fail; and an alert may not reach an authorized responder. The prototype has not been shown to integrate with official emergency dispatch. Thresholds, hardware mounting, false-alarm rates, data retention, and security require further work. These limitations are central findings, not footnotes.

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CHAPTER 6

6.4 Safety and Ethical Considerations

False alerts can consume responder attention, while missed events can create dangerous reassurance. The system should communicate uncertainty, minimize personal data, secure access, and preserve an audit trail. It should never replace official emergency services or tell users that help is on the way unless delivery and acknowledgment are actually confirmed.

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> *“Technology earns trust when it is honest about what it knows, what it does not know, and what it has actually done.”*

CHAPTER 6

6.5 Future Work: Dataset

Collect representative, consented motion data across devices, mounting positions, road conditions, and ordinary driving maneuvers. Establish reliable labels and separate training, validation, and test sets. Report class balance and uncertainty. Any field data collection must be designed with privacy, participant safety, and institutional approval in mind.

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CHAPTER 6

6.6 Future Work: Model Calibration

Compare a transparent threshold baseline with statistical or machine-learning approaches. Evaluate precision, recall, false alarms per operating hour, missed-event rate, and calibration. Select operating thresholds according to the real costs of false positives and false negatives. A more complex model should only be adopted if it offers measurable and reproducible benefit.

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CHAPTER 6

6.7 Future Work: Connectivity

Design resilient delivery with retry policies, idempotency, offline buffering, and delivery receipts. Clearly distinguish queued, transmitted, accepted, acknowledged, and resolved states. Test poor connectivity and service outages. Integration with official emergency services requires formal authorization and must follow the relevant operational and legal requirements.

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CHAPTER 6

6.8 Future Work: Security

Conduct a threat model covering device theft, spoofed locations, replayed alerts, account compromise, API abuse, and data leakage. Add role-based access control, secure transport, secret management, retention limits, backups, and security testing. Consider how incident records can be corrected without erasing the audit history.

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CHAPTER 6

6.9 Future Work: Human Factors

Evaluate alert language and confirmation timing with representative users. Use plain language and avoid overconfident wording. Include accessible design and consider the needs of people under stress, people with disabilities, and people unfamiliar with the interface. A clear status indicator should be understandable without relying on colour alone.

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CHAPTER 6

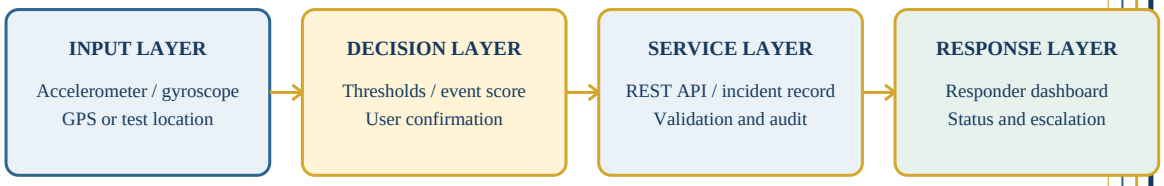
6.10 Deployment Readiness

Before deployment, require documented hazard analysis, validated sensor behavior, security review, operational ownership, responder training, incident escalation policy, monitoring, support, and an approved integration path. A successful classroom demonstration is a necessary learning milestone but not sufficient evidence for deployment in a safety-critical environment.

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Logical architecture proposed in the project; deployment components must be tested independently.

Figure 6.10 Deployment Readiness — conceptual design view.

CHAPTER 6

6.11 Social Impact Framework

Impact evaluation should measure process outcomes before claiming societal outcomes. Candidate measures include event-to-alert time, location completeness, responder acknowledgment time, duplicate rate, false-alarm burden, and user comprehension. Only a well-designed comparative study can support stronger claims about response outcomes. Benefits should be reported alongside risks and limitations.

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CHAPTER 6

6.12 Final Reflection

The most important lesson is that automation carries responsibility along with speed. A system that detects a possible accident must also explain uncertainty, protect sensitive information, and respect the people who make real emergency decisions. The next phase is not simply adding more features; it is producing stronger evidence, safer failure behavior, and accountable collaboration.

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